ORION-Class Torch Ship

Systems, Reactor Physics, and Programme Plan

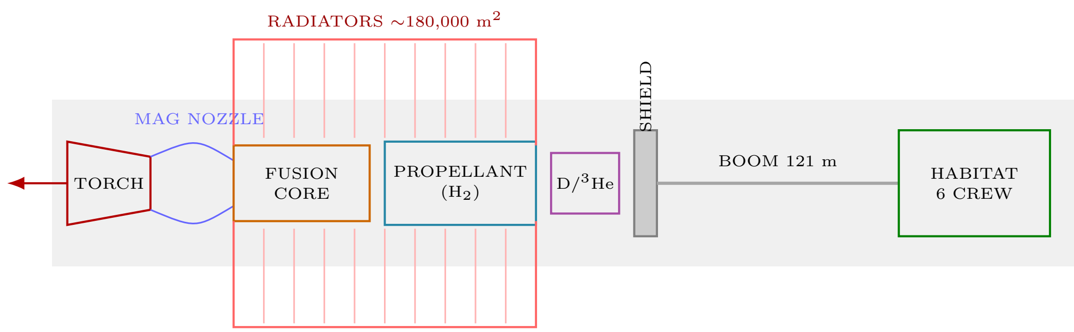
Technical reference, revision 2 — with the fusion plant worked from first principles.

Contents

1 The ship at a glance

ORION is a ~650 t fusion-torch vessel that crosses to Mars in about six days at $0.30\,g_0$ sustained acceleration. The design is driven by three hard constraints, discovered in that order:

1. **Drive specific power** sets the top speed. Before heat or radiation bind, the *mass closure loop* diverges: too much acceleration and no finite ship exists.
2. **The radiator sets the cruise, not the engine.** Rejecting waste heat by T^4 radiation alone demands enormous area.
3. **Fast transit is itself radiation shielding.** A six-day crossing cuts galactic-cosmic-ray dose ~40× versus a Hohmann transfer.



2 The reactor: what it burns and how it works

2.1 Why fusion, and which reaction

Four candidate reactions matter. The choice is a three-way fight between *how hard it is to ignite*, *how many neutrons it throws*, and *whether the energy comes out as charged particles you can use directly.*

Reaction	Products	Energy	Neutron share	Ignition T
D + T → ^{4}He + n	3.5 + 14.1 MeV	17.6 MeV	**80%**	~15 keV
D + D → (two branches)	p+T / n+^{3}He	3.7 / 3.3 MeV	~50%	~50 keV
D + ^{3}He → ^{4}He + p	**3.6 + 14.7 MeV**	**18.3 MeV**	**~0%***	~60 keV
p + ^{11}B → 3 ^{4}He	8.7 MeV	8.7 MeV	0%	~150 keV

*nominally — see §2.5, the neutron caveat.

ORION burns deuterium and helium-3.

$$\mathrm{D} + {}^3\mathrm{He} \longrightarrow {}^4\mathrm{He}\,(3.6\,\mathrm{MeV}) \;+\; \mathrm{p}\,(14.7\,\mathrm{MeV}), \qquad Q = 18.3\,\mathrm{MeV}. \tag{1}$$

The decisive property is that **both products are electrically charged**. A charged exhaust can be steered by a magnetic nozzle (so it becomes *thrust* directly), and charged particles can be decelerated against an electric field to recover power *without a heat engine.* Neither is possible with D–T, whose energy leaves as 14.1 MeV neutrons: neutrons cannot be magnetically steered, cannot be directly converted, and activate the structure. A D–T torch would be a neutron lamp bolted to a heat exchanger.

The price is difficulty. Specific energy is

$$\frac{Q}{m_{\rm D} + m_{^3{\rm He}}} = \frac{2.932 \times 10^{-12}\,{\rm J}}{8.353 \times 10^{-27}\,{\rm kg}} = 3.51 \times 10^{14}\,{\rm J/kg}, \tag{2}$$

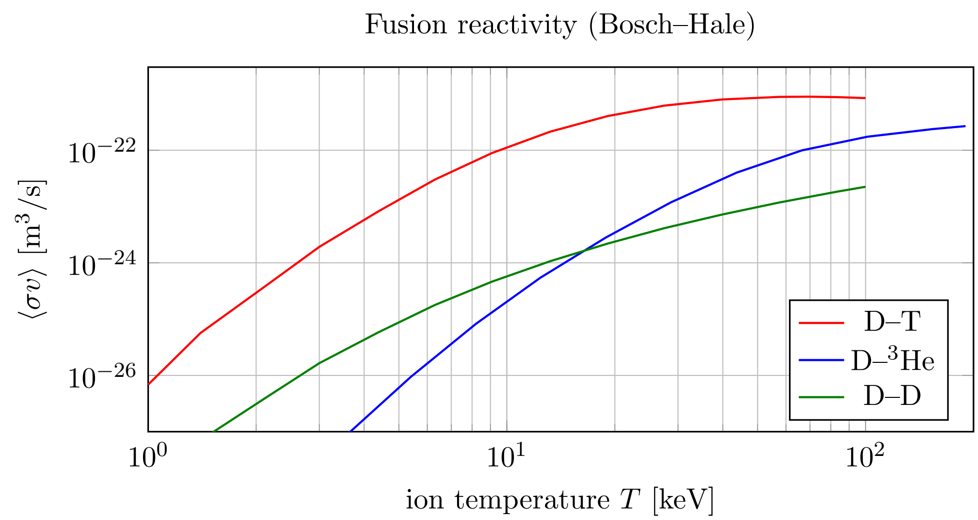
about ten million times chemical fuel — but it must be lit at hundreds of millions of kelvin.

2.2 Reactivity: the price of going aneutronic

The fusion power density depends on the *reactivity* $\langle\sigma v\rangle$, the velocity-averaged cross-section:

$$P_{\rm fus} = n_1 n_2 \langle\sigma v\rangle\, Q \qquad [{\rm W/m^3}]. \tag{3}$$

The curves below are computed from the Bosch–Hale parametrization (not sketched):



At its peak D–T reaches 8.9×10^{-22} m^3/s near 67 keV; D–^{3}He manages only 2.7×10^{-22} and needs $\sim$190 keV to get there. **D–^{3}He is roughly $3\times$ less reactive and needs $3\times$ the temperature.** That is exactly what we pay for a charged-particle exhaust, and it is why the plant runs at $T_{\rm ion} \approx$ 10–15 keV during ramp-up but must be driven far hotter to reach full gain.

2.3 Ignition: the Lawson criterion

Fusion self-sustains when the power deposited by charged products exceeds losses. In the standard form, the *triple product* must exceed a threshold:

$$\boxed{n\,T\,\tau_E \gtrsim \text{const.}} \tag{4}$$

where n is density, T temperature, τ_E the energy confinement time. Approximate requirements:

Fuel	$nT\tau_E$ [keV·s/m^3]	optimum T
D–T	$\sim 3 \times 10^{21}$	15 keV
D–^{3}He	$\sim 1 \times 10^{23}$	60 keV
p–^{11}B	$\sim 3 \times 10^{24}$	150 keV

D–^{3}He is $\sim$30$\times$ harder to ignite than D–T. This is the single fictional concession the setting makes: ORION's plant achieves a D–^{3}He triple product that real machines have not.

The **gain** Q tracked on the reactor panel is

$$Q = \frac{P_{\text{fusion}}}{P_{\text{external heating}}}, \tag{5}$$

so $Q = 1$ is breakeven and $Q \to \infty$ is full ignition. The simulator treats $Q > 5$ as *self-sustaining*: at that point the RF heaters back off and the burn carries itself. That threshold is why the ignition sequence has a distinct IGNITE→ASCENT transition.

2.4 Confinement and the magnets

The plasma is held off the walls by a toroidal magnetic field. Two numbers matter.

Plasma beta — the ratio of plasma pressure to magnetic pressure:

$$\beta = \frac{2\mu_0 nkT}{B^2}. \tag{6}$$

Confinement fails above a critical β, so higher B permits higher density and thus more power. This is why the machine wants 8 T.

Stored magnetic energy — the reason a quench is catastrophic:

$$u = \frac{B^2}{2\mu_0}, \qquad E_{\text{coil}} = \frac{B^2}{2\mu_0}V. \tag{7}$$

At $B = 8\,\text{T}$, $u = 25.5\,\text{MJ/m}^3$. With a field volume of $\sim 12\,\text{m}^3$ the coils store

$$E_{\text{coil}} \approx 306\,\text{MJ}, \tag{8}$$

consistent with the simulator's 4 MW charging over a 75-second ramp ($4 \times 10^6 \times 75 = 300$ MJ). **That is the energy of ~70 kg of TNT stored in the magnets.**

2.5 The neutron caveat — why we still carry a shield

D–^{3}He is called "aneutronic," but a deuterium plasma inevitably burns *itself*:

$$\text{D} + \text{D} \to \text{n}\,(2.45\,\text{MeV}) + {}^3\text{He} \qquad (\sim 50\%\text{ of D–D events}). \tag{9}$$

These side reactions typically put a few percent of total power into neutrons. Small compared to D–T's 80%, but not zero — which is precisely why ORION still needs the 121 m boom and the 16 cm shadow shield derived in the shielding analysis. **The fuel choice shrinks the shield; it does not delete it.**

2.6 Fuel supply — a plot hook that is also physics

Deuterium is trivial: 1 atom in 6400 of hydrogen, so seawater is an inexhaustible source. **Helium-3 is the problem.** It is essentially absent on Earth (the geomagnetic field deflects the solar wind that makes it). The two real reservoirs are:

- **Lunar regolith** — implanted by 4 billion years of solar wind, at ~10–20 parts per *billion*. Extracting one tonne means processing $\sim 10^8$ tonnes of soil.
- **Gas giant atmospheres** — Jupiter and Saturn are rich in it, but require aerocapture mining.

ORION burns ~27 grams of fuel per second (§3.2). That sounds trivial — but the ^{3}He half of it is the most expensive substance in the solar system, and this alone explains why a civilisation with torch drives would care intensely about the Moon and the outer planets.

2.7 Startup sequence — the physics behind each stage

Stage	Physics	Gate to advance	Failure mode
PUMPDOWN	evacuate chamber, $p \propto e^{-t/\tau}$	$p < 10^{-3}$ Pa	impurities quench burn
FIELD	charge coils, $E = \frac{B^2}{2\mu_0} V$	$I = 50$ kA, $B = 8$ T	**coil quench**
HEAT	gas puff + RF heating	$T_i \geq 10$ keV, $n \geq 1.0$	density disruption
IGNITE	gain climbs, $Q = P_{\text{fus}}/P_{\text{ext}}$	$Q \geq 5$	vacuum/field loss
ASCENT	thermal power ramps to 50 MW	$P_{\text{th}} = 50$ MW	coolant overtemp
ONLINE	20 MW$_e$ steady	—	scram on overtemp

Quench deserves its own paragraph, because it is the reason cooling must precede ignition. A superconductor carries current with zero resistance only below a critical surface in $(T,\, B,\, J)$. If any part of the winding warms past T_c, that spot goes *normal* — it acquires resistance, ohmic heating I^2R dumps energy locally, the normal zone propagates, and the coil tries to dissipate its *entire* 306 MJ into a small volume of metal. The magnet destroys itself. Hence the simulator's rule: no coolant flow during FIELD $\Rightarrow$ coil temperature climbs $\Rightarrow$ quench abort at 40 K.

2.8 Power conversion

Because the products are charged, two paths exist:

$$\text{Thermal cycle:} \quad \eta \approx 0.35\text{–}0.45 \quad (\text{blanket} \to \text{turbine}) \tag{10}$$

$$\text{Direct conversion:} \quad \eta \approx 0.6\text{–}0.8 \quad (\text{decelerate ions against a field}) \tag{11}$$

ORION's *housekeeping* reactors use a thermal cycle at $\eta = 0.40$ (50 MW$_{\text{th}} \to$ 20 MW$_e$, 30 MW waste heat — which is exactly the heat load the coolant loop must reject). The *main torch* needs no conversion at all: the plasma is simply exhausted through a magnetic nozzle. This is why the two loops are modelled separately.

3 The torch: mass augmentation (a system we have not yet modelled)

3.1 From fusion products to thrust

If the plasma were exhausted directly, its velocity would follow from the reaction energy:

$$v_{\text{prod}} = \sqrt{\frac{2Q}{m_{\text{D}} + m_{^3\text{He}}}} = \sqrt{\frac{2(2.932 \times 10^{-12})}{8.353 \times 10^{-27}}} = 2.65 \times 10^7 \text{ m/s}, \tag{12}$$

which is 8.8% of light speed — an I_{sp} of 2.7×10^6 s.

But ORION's design v_{ex} is 9.81×10^6 m/s ($I_{sp} = 10^6$ s) — deliberately *lower*. Why throw away performance?

3.2 The reason: thrust is bought by diluting the exhaust

For a *fixed* jet power P,

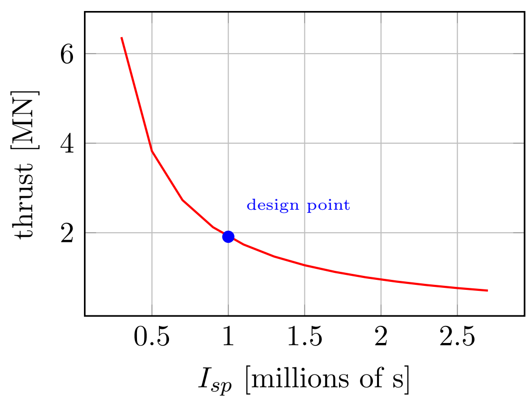
$$P = \tfrac{1}{2}\dot{m}\, v_{\text{ex}}^2 = \tfrac{1}{2} F\, v_{\text{ex}} \qquad \Longrightarrow \qquad \boxed{F = \frac{2P}{v_{\text{ex}}}} \tag{13}$$

Thrust is *inversely* proportional to exhaust velocity. A pure-fusion-product exhaust is magnificently efficient and nearly useless for pushing a 650-tonne ship: you would get a third of the thrust. So the drive **injects inert propellant (hydrogen) into the plasma stream**, which shares the energy, slows the exhaust, and multiplies the thrust. Energy conservation fixes the mixing ratio exactly:

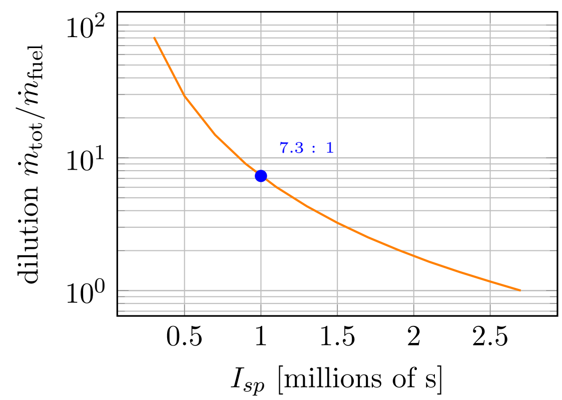
$$\tfrac{1}{2}\dot{m}_{\text{tot}} v_{\text{ex}}^2 = \tfrac{1}{2}\dot{m}_{\text{fuel}} v_{\text{prod}}^2 \qquad \Longrightarrow \qquad \boxed{\frac{\dot{m}_{\text{tot}}}{\dot{m}_{\text{fuel}}} = \left(\frac{v_{\text{prod}}}{v_{\text{ex}}}\right)^2} \tag{14}$$

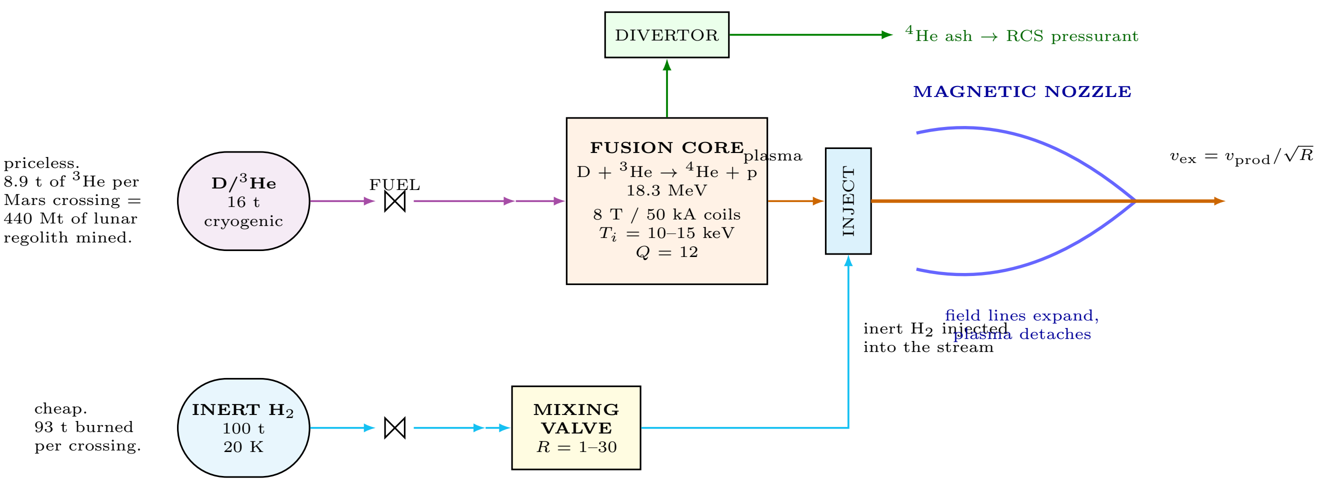
At the design point: $(2.65 \times 10^7 / 9.81 \times 10^6)^2 = \mathbf{7.30}$. And independently, from the mass flows: $\dot{m}_{\text{tot}} = F/v_{\text{ex}} = 0.195\,\text{kg/s}$, while the fuel burn needed to make 9.37 TW is $\dot{m}_{\text{fuel}} = P/(Q/m) = 9.37 \times 10^{12}/3.51 \times 10^{14} = 0.0267\,\text{kg/s}$. Ratio: $0.195/0.0267 = \mathbf{7.30}$. *The two derivations agree exactly* — the model is self-consistent.

Fixed 9.37 TW: thrust vs. I_{sp}



Propellant added per unit fuel





THE MIXING VALVE. Energy conservation fixes everything:
$\tfrac{1}{2}\dot{m}_{\text{tot}} v_{\text{ex}}^2 = \tfrac{1}{2}\dot{m}_{\text{fuel}} v_{\text{prod}}^2 \;\Rightarrow\; R \equiv \dot{m}_{\text{tot}}/\dot{m}_{\text{fuel}} = (v_{\text{prod}}/v_{\text{ex}})^2 \;\Rightarrow\; v_{\text{ex}} = v_{\text{prod}}/\sqrt{R}$

At fixed jet power P: $F = 2P/v_{\text{ex}} \propto \sqrt{R}$, and $a_{\max} = 2P_{\max}\sqrt{R}/(m\, v_{\text{prod}})$.

Waste heat is $f \cdot P$ — it does not depend on R at all.
⇒ Diluting the exhaust buys 5.5× the thrust for *zero* thermal cost and *zero* extra fusion fuel. You pay purely in inert hydrogen.

R	v_{ex} [m/s]	I_{sp} [Ms]	$a_{\max}$ [g]	radiator
1.0	2.65×10^7	2.70	0.178	**1690 K**
7.3	9.81×10^6	1.00	0.480	**1690 K**
30.0	4.84×10^6	0.49	0.973	**1690 K**

Radiator temperature is identical at every mixture.

3.3 The mixing valve is the most important control in the ship

R is a **real-time pilot control** that trades I_{sp} against thrust at constant reactor power. Since $v_{\text{ex}} = v_{\text{prod}}/\sqrt{R}$ and $F = 2P/v_{\text{ex}}$, the *maximum attainable acceleration* scales as

$$a_{\max} = \frac{2P_{\max}\sqrt{R}}{m\,v_{\text{prod}}} \;\propto\; \sqrt{R}. \tag{15}$$

The simulator's own output, at $P_{\max} = 15\,\text{TW}$ on a 650 t ship:

R	v_{ex} [m/s]	I_{sp} [Ms]	$a_{\max}$ [g]	$\dot{m}_{\text{fuel}}$	**radiator**
1.0	2.650×10^7	2.702	0.178	0.0427	**1690 K**
2.0	1.874×10^7	1.911	0.251	0.0427	**1690 K**
4.0	1.325×10^7	1.351	0.355	0.0427	**1690 K**
7.3	9.808×10^6	**1.000**	**0.480**	0.0427	**1690 K**
15.0	6.842×10^6	0.698	0.688	0.0427	**1690 K**
30.0	4.838×10^6	0.493	0.973	0.0427	**1690 K**

Read the last two columns. Across a 30× change in mixture, the fusion fuel burn is *identical* and the radiator temperature is *identical*. Waste heat is $f \cdot P$ — it depends on the *power*, and the mixing valve does not change the power. So:

> **Diluting the exhaust buys 5.5× the thrust for zero thermal cost and zero extra fusion fuel.**
> **You pay for it purely in inert hydrogen.**

This is the cleanest trade in the vehicle:

- **Rich** ($R \to 30$): nearly 1 g available. Hard manoeuvring, emergency burns. Drinks the H_2 tank.
- **Lean** ($R \to 1$): I_{sp} of 2.7 million seconds, but only $0.18\,g_0$. Economical cruise; enormous Δv reserve.
- If the commanded acceleration exceeds what the plant can supply, **the pilot's remedy is to richen the mixture** — not to demand more power.

Two tanks, depleting at different rates: **100 t of cheap inert hydrogen** and **16 t of priceless D/^{3}He**. A Mars crossing at the design point burns 93.0 t of the former and 14.8 t of the latter.

Fuel for a Mars run: $0.0267\,\text{kg/s} \times 5.53 \times 10^5\,\text{s} =$ **14.8 t** of D/^{3}He, of which 3/5 by mass — **8.9 t** — is ^{3}He. At a lunar regolith concentration of 20 parts per billion, that single crossing represents **~440 million tonnes of regolith processed.** The propellant is cheap; the fuel is an industrial programme.

4 Guidance avionics (after Apollo)

The Apollo CSM solved these problems, and its failure modes are the interesting ones. ORION copies the architecture.

4.1 The stable platform, and gimbal lock

The IMU is a gyro-stabilised platform on three gimbals, holding an inertial reference (the REFSMMAT) that all guidance is measured against. Gimbals have a fatal geometry: when the **middle gimbal** rotates far enough, the outer and inner axes become nearly parallel, the platform loses a degree of freedom, and it **tumbles**. The reference is destroyed.

Apollo lit its GIMBAL LOCK warning at ±70° of middle gimbal. ORION uses the same numbers:

Middle gimbal	State	Consequence
$< 70°$	normal	—
70°–85°	GIMBAL LOCK warning	manoeuvre away *now*
$> 85°$	**locked**	platform tumbles; reference **lost**

4.2 Why there is a second, worse attitude reference

The **GDC** integrates the rate gyros instead of holding a platform. It drifts 20× faster (18 vs. 0.9 arcmin/hour) — but **it has no gimbals, so it cannot lock**. The manoeuvre that destroys the IMU leaves the GDC intact. One button (GDC ALIGN) copies the IMU's reference across while you still have it. This is redundancy through *dissimilarity*: not two of the same box, but two boxes that fail differently.

4.3 Platform alignment (P52)

Gyros drift, so the platform must be re-aligned against stars. The sequence, and its gates:

Phase	Action	Gate
COARSE ALIGN	torque platform roughly into place	rates $< 0.05°/s$
MARK 1	sight first catalogue star	rates $< 0.05°/s$
MARK 2	sight second star	rates $< 0.05°/s$
STAR ANGLE DIFF	compare measured vs. catalogued separation	within limits, else **reject**
TORQUING	apply computed torquing angles	—

The **star angle difference** is the integrity check: the true angle between two catalogue stars is known exactly, so if your two marks disagree with it, you sighted the wrong star and the fix is thrown out. Fire a thruster mid-mark and you lose the mark.

Four orientation options are offered, as on Apollo:

1 **Preferred** — orient to the attitude the next burn requires.

2 **Nominal** — local vertical / local horizontal at a specified time.

3 **REFSMMAT** — keep the stored reference; null the accumulated drift only.

4 **Target** — align to the selected target's line of sight.

Alignment runs on a **fixed schedule** (default every 6 hours). At 0.9 arcmin/hour the platform accumulates ∼5.4 arcmin between alignments; past 8 arcmin the nav state is flagged DEGRADED and the computer refuses to issue a burn solution.

4.4 Reaction control — and why it is *not* a torch

Sizing an RCS thruster with the main drive's exhaust velocity is instructive. At fixed power $F = 2P/v_{\text{ex}}$, so 60 kN of attitude thrust at $v_{\text{ex}} = 9.81 \times 10^6$ m/s would demand

$$P = \tfrac{1}{2}Fv_{\text{ex}} = \tfrac{1}{2}(6 \times 10^4)(9.81 \times 10^6) = 2.9 \times 10^{11}\,\text{W} = \mathbf{294}\,\text{GW}\ \textbf{per thruster}. \tag{16}$$

A dedicated fusion plant on every corner of the ship. Worse, a fusion plasma cannot be pulsed on a 20 ms attitude blip, and a relativistic plume fired sideways would scour the vehicle. And it buys nothing: the RCS Δv budget is ∼100–200 m/s for the entire mission, where the rocket equation is indifferent to I_{sp}.

So the RCS is conventional hypergolic, pressure-fed — exactly Apollo's. MMH fuel and N_2O_4 oxidiser at O/F = 1.6, $I_{sp} \approx 300$ s, igniting on contact so there is no igniter, no pump, and thousands of restarts. Four quads of four thrusters.

Pressure feed means helium, not propellant, is the thing you run out of. High-pressure helium is regulated down and squeezes a **bladder** inside each propellant tank. Two helium bottles per quad, because one regulator failure would otherwise kill an entire axis. *A quad with full tanks and closed helium valves produces exactly zero thrust.* And the propellants freeze: N_2O_4 solidifies at 262 K, so the quads are heated continuously — an unheated quad is a dead quad, however full.

4.5 The ship makes its own pressurant

Why helium? It is inert, and it is the lightest inert gas: pressurant mass is nM with $n = PV/RT$ fixed by the tank, so low molar mass wins.

Could ^{3}He be the pressurant? Physically yes — chemically identical to ^{4}He and *lighter* ($M = 3$ vs. 4), hence 25% more mass-efficient. Economically it is madness: pressurant is vented overboard, and ^{3}He is the most expensive substance aboard.

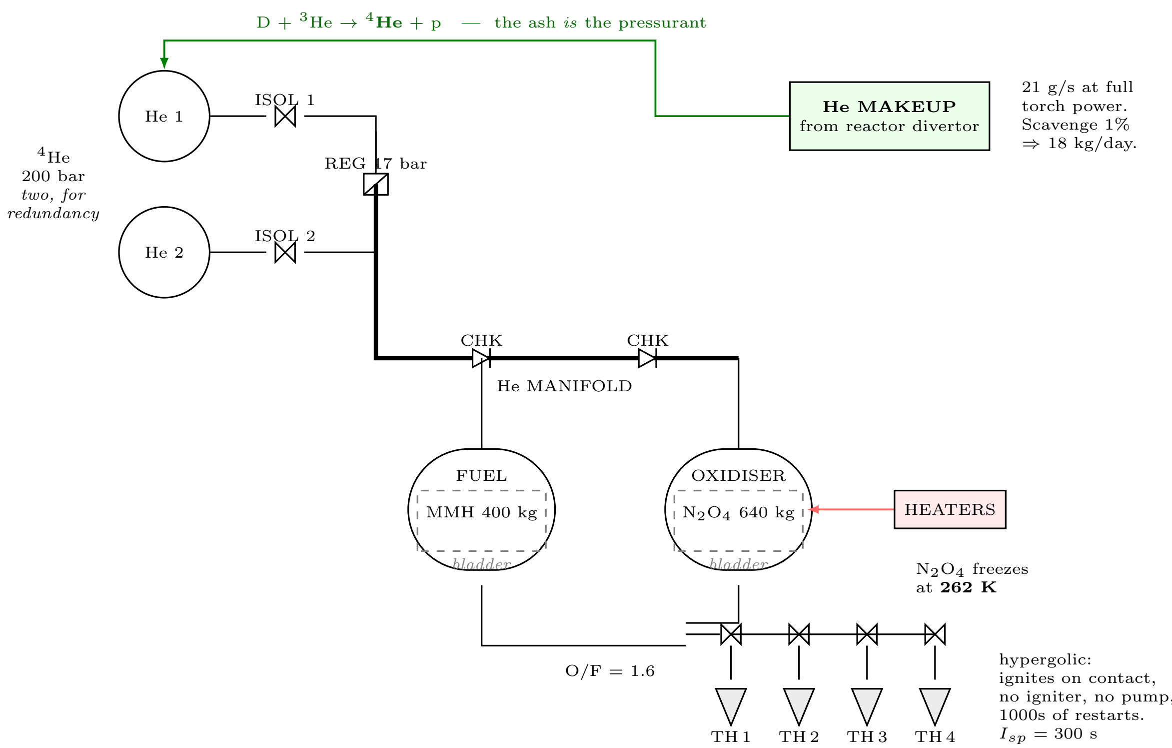
But look at the reaction again:

$$\mathrm{D} + {}^3\mathrm{He} \longrightarrow \boxed{{}^4\mathrm{He}} + \mathrm{p}. \tag{17}$$

The reactor's ash *is* helium-4. Four of the five reactant amu leave as an alpha particle. At full torch power the plant produces

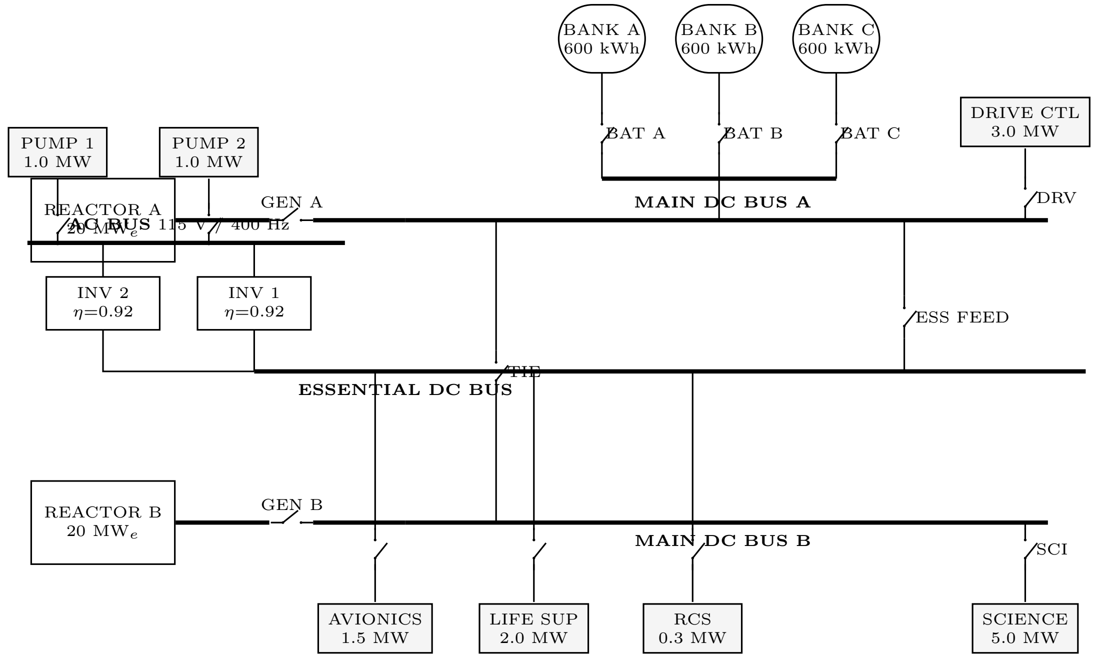
$$\dot{m}_{^4\mathrm{He}} = \tfrac{4}{5}\dot{m}_{\mathrm{fuel}} = 0.8 \times 0.0267 = 0.0214\,\mathrm{kg/s} \approx 21\,\mathrm{g/s}. \tag{18}$$

Scavenging even 1% from the divertor yields ∼18 kg/day — far more than the ∼40 kg the RCS needs. **The ship refills its own pressurant from its own exhaust.** Note the consequence: makeup only runs *while the torch is burning.* The housekeeping reactors, at 10^{-5} of the torch's power, contribute nothing. Coast for a long time while manoeuvring hard, and you can run the helium down with no way to replace it.



NOTES. (1) Helium squeezes the **bladder**; the propellant has no pump and no ullage problem.
(2) Blowdown: as propellant leaves, He expands and tank pressure falls. Below ∼20 bar, thrust collapses *with fuel still aboard.*
(3) Makeup only runs **while the torch is burning**. The housekeeping reactors (10^{-5} of torch power) contribute nothing.
(4) Four such quads (A/B/C/D), 16 thrusters total, giving full attitude *and* translation authority.

5 Electrical power (EPS)



NOTES. (1) ESS FEED closed ⇒ reactors carry the ship and the banks recharge.
(2) TIE closed ⇒ either reactor alone feeds *both* main buses.
(3) Pumps are **AC**: no inverter ⇒ no coolant flow ⇒ reactor scrams in 30 s.
(4) Bus volts sag with state of charge ($V = 700 + 300S$). $I = P/V$, so a *draining* battery raises current. ESS FEED trips at 12,000 A — load-shed or lose the bus.

The bus is a network of stores and converters. With battery state E:

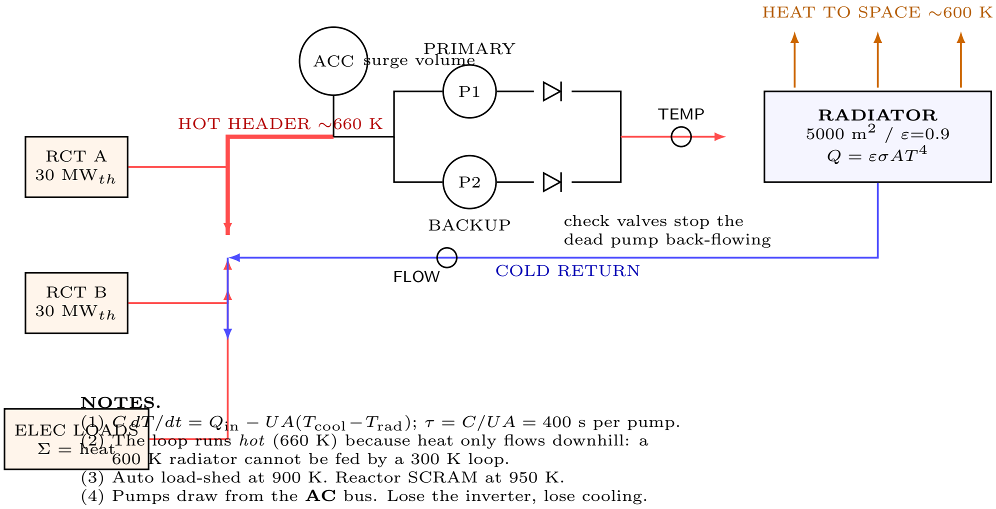
$$\frac{dE}{dt} = P_{\text{gen}} - P_{\text{load}}, \qquad 0 \le E \le E_{\max}, \qquad I = \frac{P}{V}. \tag{19}$$

Two design details drive all the gameplay:

Voltage sag. Battery terminal voltage falls with state of charge, modelled linearly as $V = 700 + 300\,S$ volts. Since $I = P/V$, *a draining battery draws more current for the same load.* Push the essential bus past its 12,000 A breaker rating during a cold start and it trips — a genuine emergent failure, not a scripted one.

Inverter losses are heat. Pumps are AC; the inverter is $\eta = 0.92$, so DC draw is P_{AC}/η and the missing 8% lands in the coolant loop. Conservation of energy means **every watt consumed anywhere becomes heat you must radiate.**

6 Thermal control



Rejection is Stefan–Boltzmann:

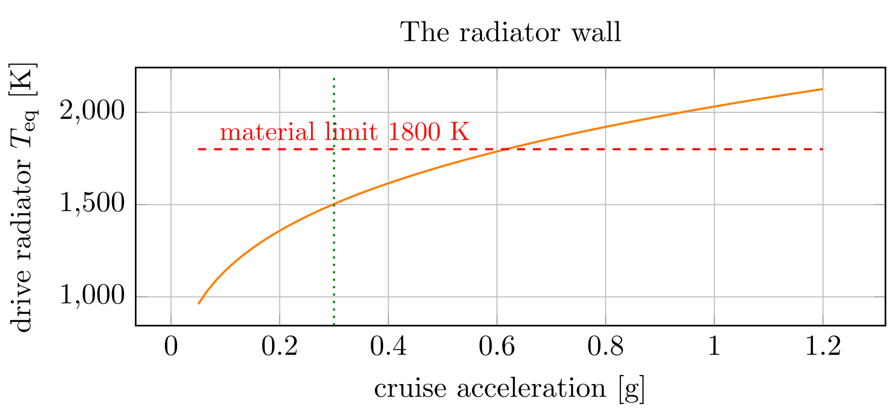
$$\boxed{Q_{\text{rad}} = \varepsilon\sigma A T^4} \qquad T_{\text{eq}} = \left(\frac{Q}{\varepsilon\sigma A}\right)^{1/4}. \tag{20}$$

The fourth power is unforgiving: doubling heat load raises radiator temperature only 19%, but halving the area demands 19% more temperature. **Radiators must run hot** — ORION's housekeeping loop sits near 650–670 K precisely so its radiator (at ∼600 K) can shed anything at all. Heat only flows downhill; a 300 K coolant loop cannot push heat into a 700 K radiator.

Transport between loops is a lumped conductance:

$$Q_{\text{pump}} = UA\,(T_{\text{cool}} - T_{\text{rad}}), \qquad C\frac{dT}{dt} = Q_{\text{in}} - Q_{\text{out}}, \qquad \tau = \frac{C}{UA}. \tag{21}$$

With $C = 2 \times 10^8$ J/K and $UA = 5 \times 10^5$ W/K per pump, $\tau = 400$ s — slow enough to fight, fast enough to kill you. **With no pump running there is no path from the loop to the radiator at all**, and the temperature diverges regardless of load shedding. That is why the pump is redundant.



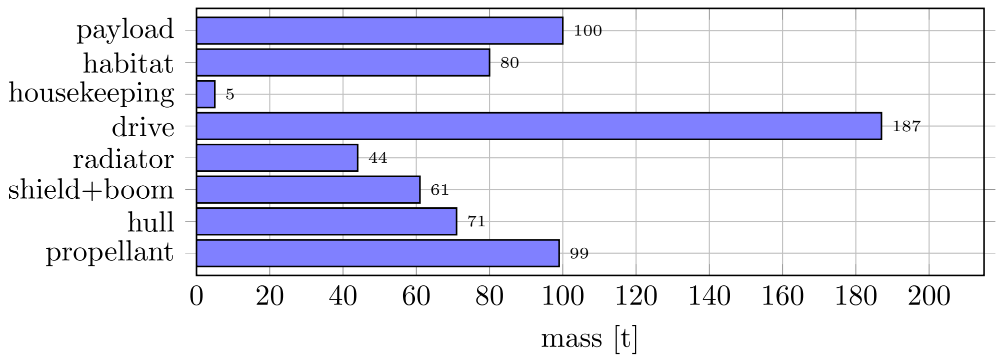
7 Mass closure

Every subsystem's mass feeds $F = ma$, so the ship is a *fixed point*, not a sum:

$$m_{\text{drive}} = \frac{\frac{1}{2} m_{\text{wet}}\, a\, v_{\text{ex}}}{(P/m)_{\text{spec}}}, \qquad m_{\text{rad}} = \frac{Q_{\text{waste}}}{\varepsilon \sigma T_{\text{lim}}^4} \rho_A, \tag{22}$$

$$m_{\text{dry}} = \big(\text{fixed} + m_{\text{drive}} + m_{\text{rad}} + m_{\text{shield}}\big)(1 + f_{\text{hull}}), \qquad m_{\text{prop}} = m_{\text{dry}} \Big(e^{\Delta v / v_{\text{ex}}} - 1\Big). \tag{23}$$

Because $m_{\text{drive}} \propto m_{\text{wet}}$, the loop can diverge. Solving iteratively at $0.30\, g_0$ with a 50 MW/kg drive gives a **646 t ship**:



The drive is 29% of the ship. **Drive specific power is the master constraint**: at 25 MW/kg no ship closes above $0.34\, g_0$; at 50 MW/kg the wall is $0.58\, g_0$; at 100 MW/kg, $0.92\, g_0$. Heat and radiation never get a chance to bind first.

8 How many nozzles?

A real design question with a decisive answer. Four arguments *for* splitting the drive, one hard constraint on *how*.

8.1 For: engine-out survivability — the strongest argument

A brachistochrone is uniquely unforgiving. Deceleration must *equal* acceleration: you spend half the trip building 814 km/s and the other half killing it. Lose thrust at the flip and the required decel is exactly a — **with less thrust you cannot stop, and you fly past Mars at a substantial fraction of a percent of light speed.**

But if you lose a nozzle *early*, you can flip early and still make it:

Nozzles	Thrust	Flip at	Trip time
4 / 4	100%	50.0% of range	6.40 d
3 / 4	75%	42.9% of range	6.91 d (+8%)
2 / 4	50%	33.3% of range	7.84 d (+22%)
1 / 4	25%	20.0% of range	9.80 d (+53%)

A single-nozzle ship has no engine-out case at all. A four-nozzle ship loses one and arrives 8% late. That alone justifies the split.

8.2 For: thrust vector control without gimballing

Gimballing a multi-gigawatt superconducting magnetic nozzle is mechanically absurd. Differential throttling gives steering for free: with four nozzles on a radius r, throttling one by 10% produces

$$\tau = \Delta F \cdot r = (0.10 \times 478\,\mathrm{kN})(5\,\mathrm{m}) = 239\,\mathrm{kN\,m}, \tag{24}$$

which is **four times** the torque of an entire RCS quad (60 kN m) — and it costs no RCS propellant, because you are burning anyway.

8.3 For: throttle turn-down

A magnetic nozzle has a *minimum* stable plasma flow; below it the plasma will not detach from the field lines. If each nozzle throttles only 40–100%, then:

Nozzles lit	Total thrust available
1	10–25%
2	20–50%
3	30–75%
4	40–100%

Continuous 10–100% coverage. A single nozzle could manage only 40–100%.

8.4 Against — and the geometry that dictates the layout

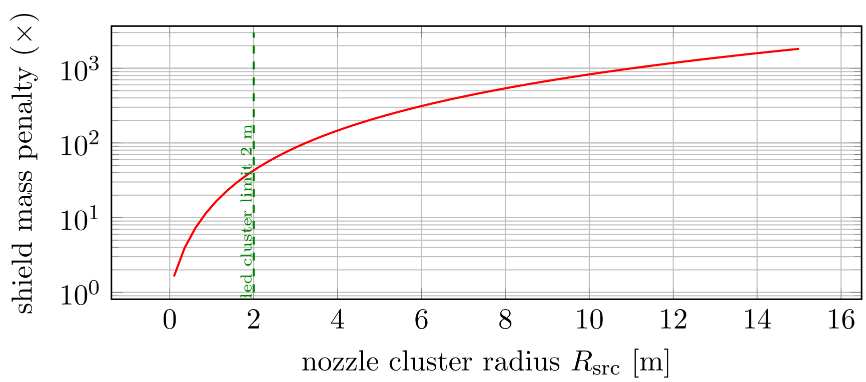
Costs are real: more coils, more cryogenics, more plasma feeds, more quench-protection circuits, and a worse surface-to-volume ratio (smaller nozzles detach plasma less efficiently).

But the **decisive** constraint is the shadow shield. The shield works because the drive is a *point* source directly behind the crew. Spread the source over a radius R_{src} and the penumbra widens the shield:

$$R_{\mathrm{shield}} = R_{\mathrm{crew}}\frac{x}{L} + R_{\mathrm{src}}\Big(1 - \frac{x}{L}\Big), \tag{25}$$

with x the shield's distance from the source and $L = 121$ m the boom. Shield *mass* scales as R^2_{shield}:

Spreading the drive destroys the shadow shield



R_{src}	Cluster span	Shield mass penalty
0 m (point)	—	1×
1 m	2 m across	14×
2 m	**4 m across**	**43×**
5 m	10 m across	221×
15 m	30 m across	1817×

8.5 Verdict

Four nozzles, tightly clustered on the centreline ($R_{\mathbf{src}} \leq 2$ m).
You buy engine-out survival, gimbal-free thrust vectoring, and a 10–100% throttle range.
You pay in coil mass and complexity — but *only if you keep them close.*
Spread them across the aft end for a prettier silhouette and the shadow shield eats the ship.

The shield penalty is the reason real nuclear-electric designs cluster their engines, and it is a satisfying constraint: the *radiation* problem, not the propulsion problem, dictates the engine layout.

9 Navigation summary

Detailed derivations are in the companion document; the operational summary:

Route	Mode	Δv	Time	Note
Earth→Moon	torch $0.30g_0$	67 km/s	**6.3 h**	straight line
Earth→Moon	ballistic (TLI)	3.14 km/s	5.0 d	Apollo-like
Earth→Mars	torch $0.30g_0$	1627 km/s	**6.4 d**	brachistochrone
Earth→Mars	Hohmann	5.59 km/s	259 d	780 d synodic wait

Guidance uses a universal-variable **Lambert** solver (Stumpff functions, bisection on z) for arbitrary flight times, and an iterative intercept solver for the brachistochrone, since the target moves during transit. Note that $\Delta\nu = 180°$ is a genuine *singularity* of Lambert's problem — the transfer plane is undefined — and the solver correctly refuses it.

10 Systems to build next

1. **Direct energy conversion** — charged-product power recovery at $\eta \approx 0.7$, as an alternative to the thermal cycle; changes the waste-heat budget dramatically.
2. **Magnetic nozzle** — throat field strength, detachment efficiency, and a failure mode where the plasma fails to detach and scours the nozzle.
3. **Fuel cryogenics** — $D/^3He$ storage, boil-off, and the tritium bred from D–D side reactions.
4. **Neutron activation** — structure becomes radioactive over mission life; a maintenance-and-dose mechanic.
5. **Life support & consumables** — $O_2/CO_2/H_2O$, already prototyped in the LCARS build.
6. **Optics subsystem** — sextant and scanning telescope, so star marks are *flown* rather than assumed.
7. **Comms** — Friis link budget, light-lag; prototyped, not yet ported.

11 Programme plan

Decision taken: build as an **Orbiter 2024 addon** (MIT-licensed, n-body Newtonian, VSOP87/ELP2000 ephemerides). Orbiter supplies orbital mechanics and rendering; we supply the ship.

Architectural rule: `ShipCore` has *zero* Orbiter dependencies. It compiles and unit-tests standalone; `Orion.cpp` (a `VESSEL4` subclass) is thin glue with no physics in it. We never reimplement orbital mechanics.

Phase	Status	Content
0. Physics core	**done**	Propulsion, thermal, shielding, mass closure — validated in Python
1. Interactive prototypes	**done**	Browser consoles: cold-start, systems, LCARS, GN&C
2. Navigation	**done**	Lambert (0.00 km miss verified), brachistochrone, Hohmann windows
3. `ShipCore` in C++	**done**	Reactor/EPS/ECS, 6/6 unit tests passing, engine-agnostic
4. Orbiter vessel	*in progress*	`VESSEL4` glue, fixed-timestep substepping, scenario persistence
5. Nav MFD	*in progress*	Lambert + brachistochrone targeting on Orbiter's state vector
6. 2D panel	see below	Clickable breakers, reactor gauges — the *Reentry* feel
7. Mass augmentation	**done**	Mixing valve in `ShipCore`; 9/9 tests passing
8. Avionics	**done**	IMU + gimbal lock, P52 (4 options, scheduled), GDC, RCS quads
9. 2D panel	**next**	Clickable breakers, reactor gauges, FDAI
10. Virtual cockpit	planned	3D panel, mesh, switches

Known hazards for phase 4–5

- Orbiter's `simdt` reaches hundreds of seconds under time acceleration. **Never** feed it to an ODE integrator — accumulate and substep at fixed 0.1 s.
- The addon DLL is *not* unloaded between scenarios; global mutable state leaks into the next flight.
- Orbiter's `Isp` argument to `CreateThruster` is exhaust velocity in **m/s**, not seconds. Pass 9.81×10^6 directly.
- Thruster handles must be created in `clbkSetClassCaps`; touching them earlier is a crash-to-desktop.
- The mixing valve changes v_{ex} *at runtime*, so `Orion.cpp` must call `SetThrusterIsp()` every step — a thruster's I_{sp} is not fixed once created.